StreamScale

Kafka → transactional PostgreSQL · scaling experiment

75 measured runs · 5 modes × 5 scenarios

ML-predictive vs its ML-reactive ablation and the M7 controllers

All acknowledged IDs reconciled before/after restart

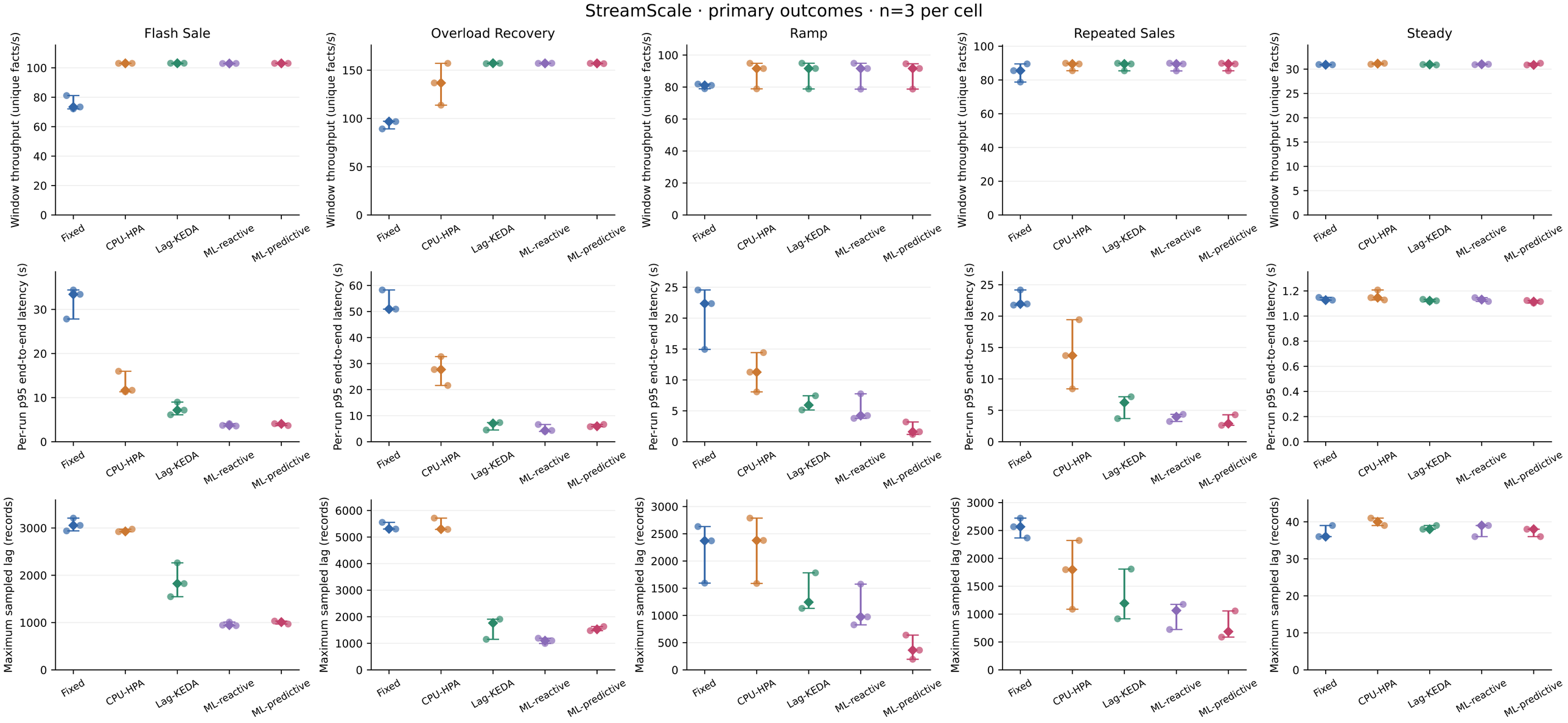
Frozen rates: 32/s steady · 250/s surge

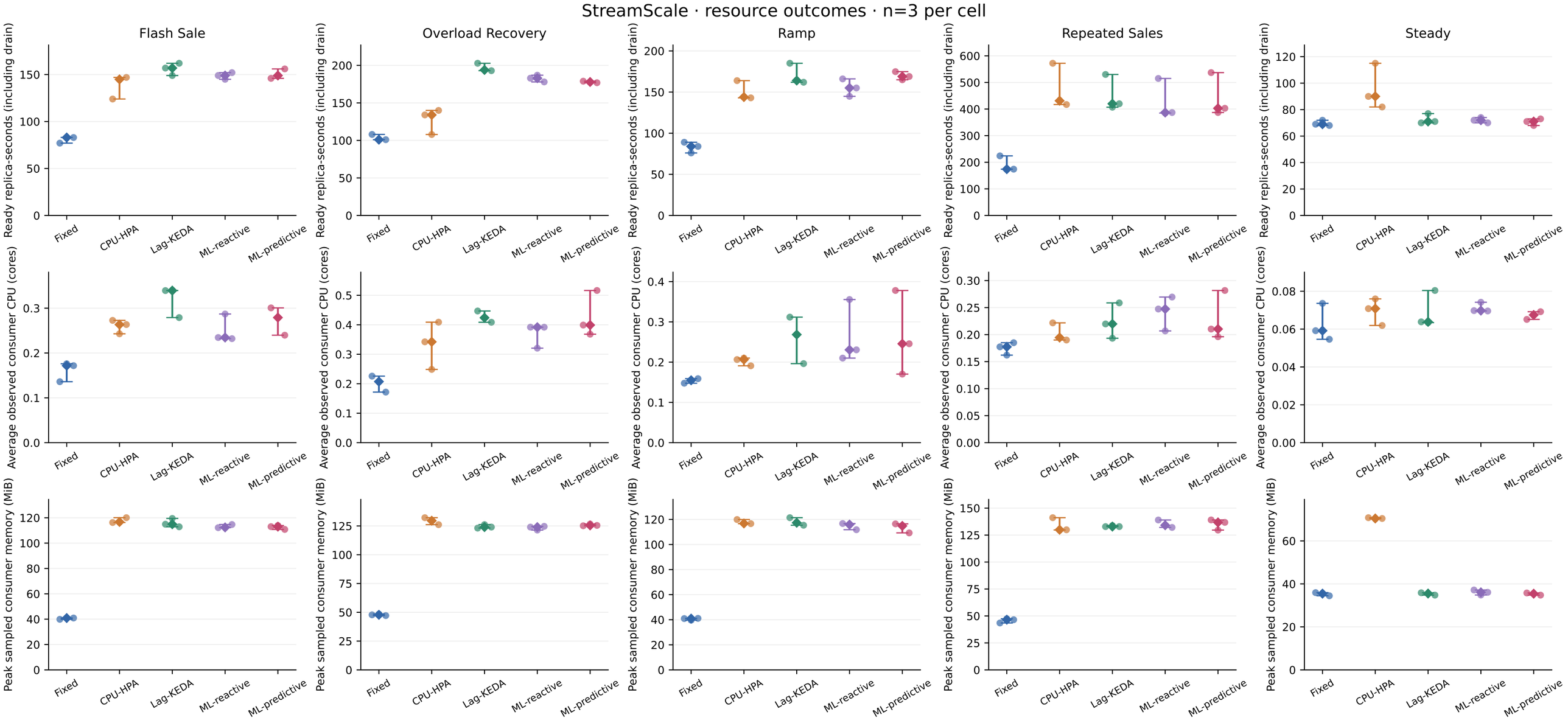
Dots = repetitions; diamonds = median; whiskers = observed range

SQL p95 includes the drained tail; throughput uses the publishing window

One shared kind node, small sample, sampled resources; descriptive results

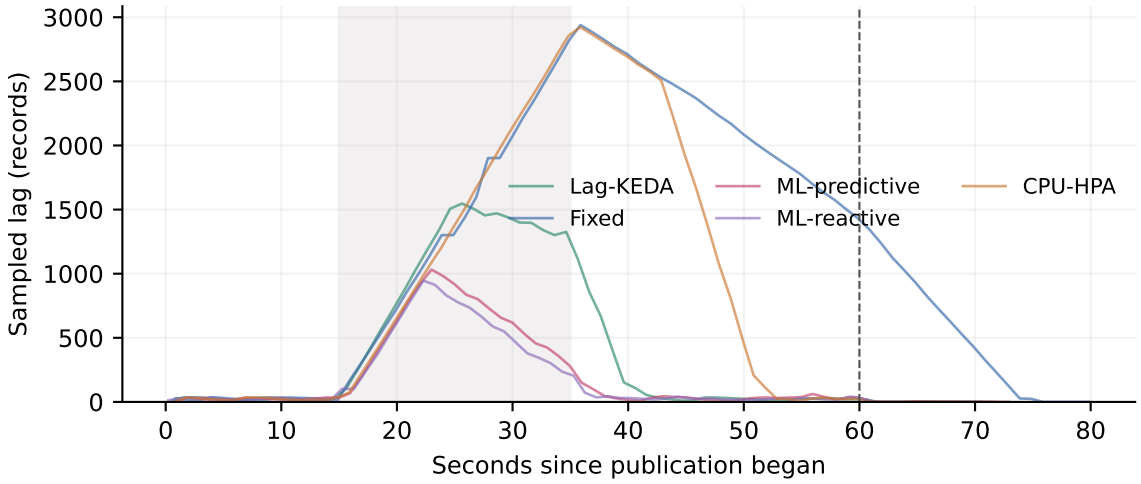
See report.md, summary.csv, frozen-protocol.json and provenance.json

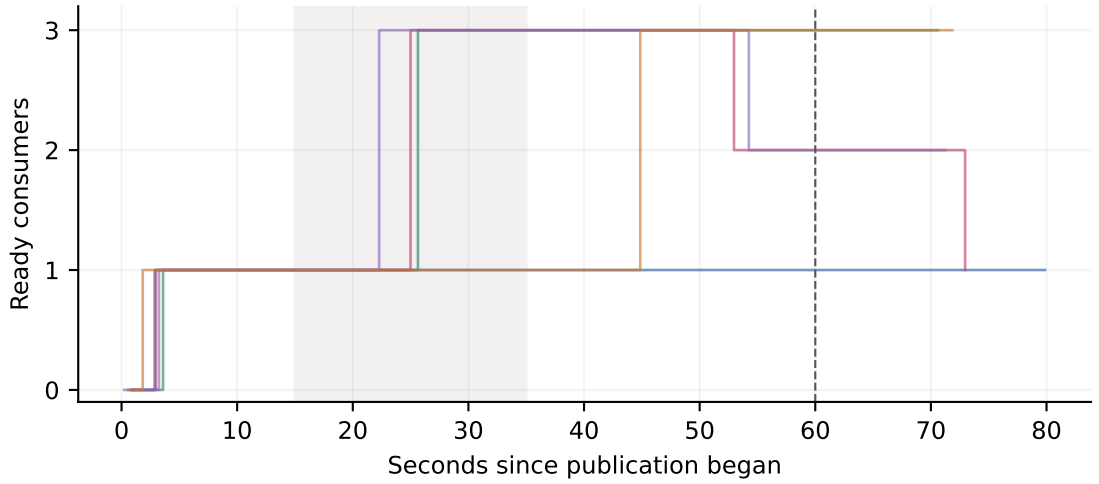

StreamScale · primary outcomes · n=3 per cell
Flash Sale
Overload Recovery
Ramp
Repeated Sales
Steady
Window throughput (unique facts/s)
Per-run p95 end-to-end latency (s)
Maximum sampled lag (records)
Fixed
CPU-HPA
Lag-KEDA
ML-reactive
ML-predictive


StreamScale · resource outcomes · n=3 per cell
Flash Sale
Overload Recovery
Ramp
Repeated Sales
Steady
Ready replica-seconds (including drain)
Average observed consumer CPU (cores)
Peak sampled consumer memory (MiB)
Fixed
CPU-HPA
Lag-KEDA
ML-reactive
ML-predictive

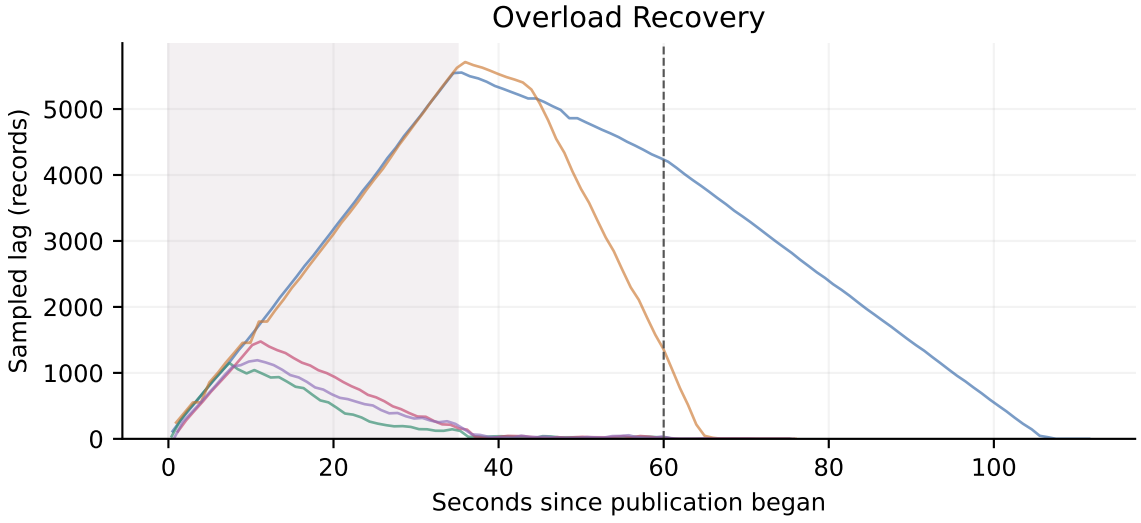
StreamScale · measured trajectories · repetition 1 of each shape

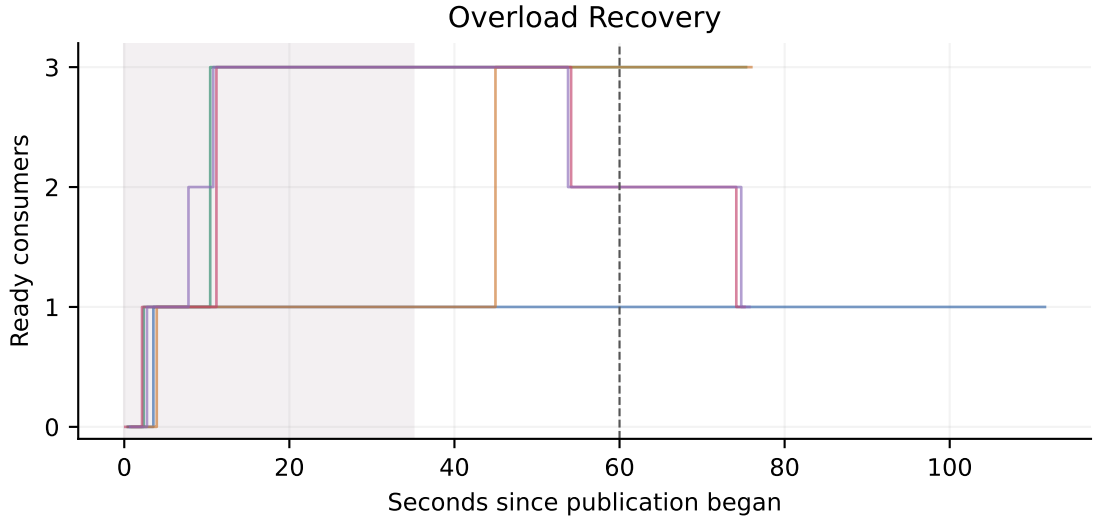
Flash Sale



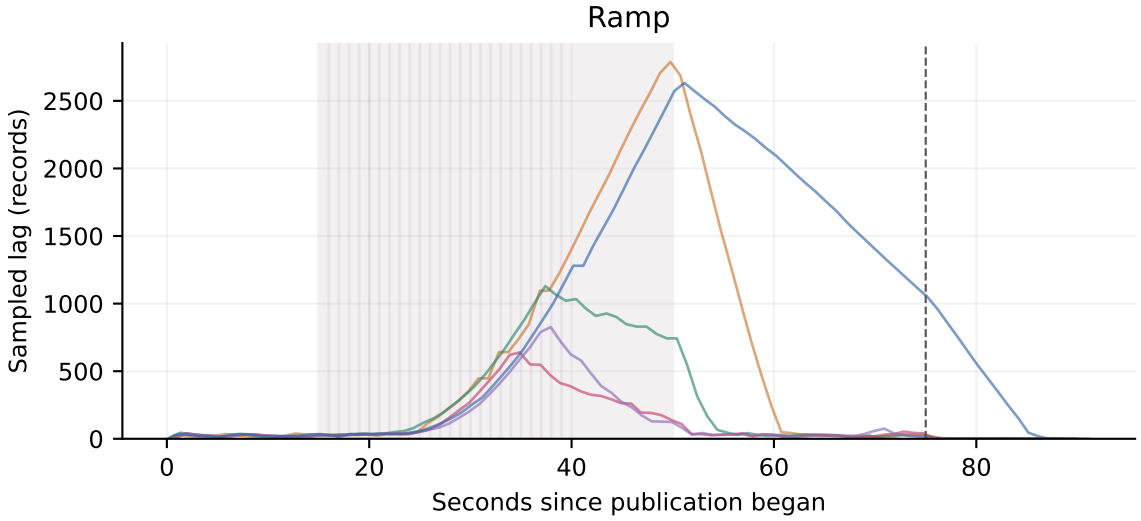


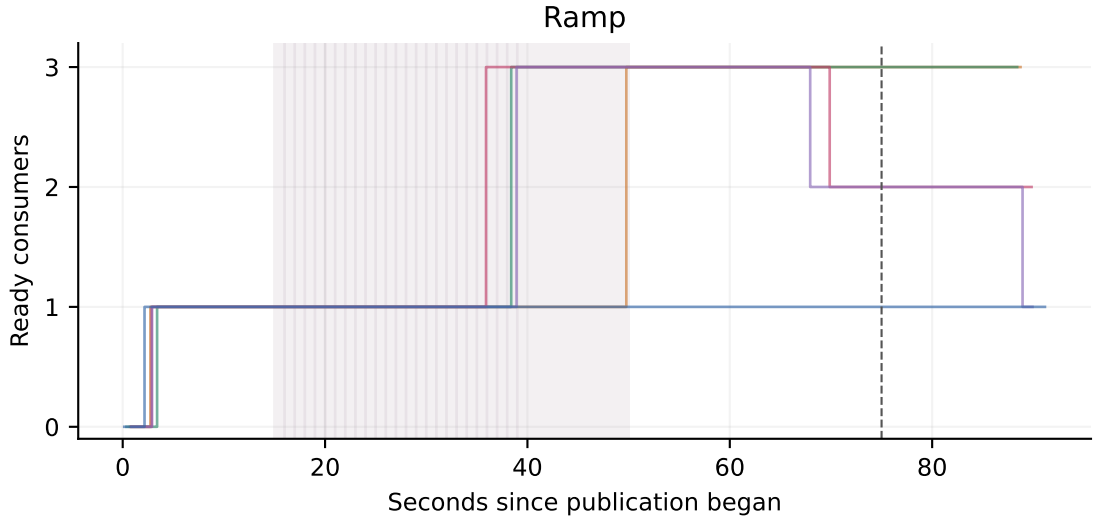
Overload Recovery



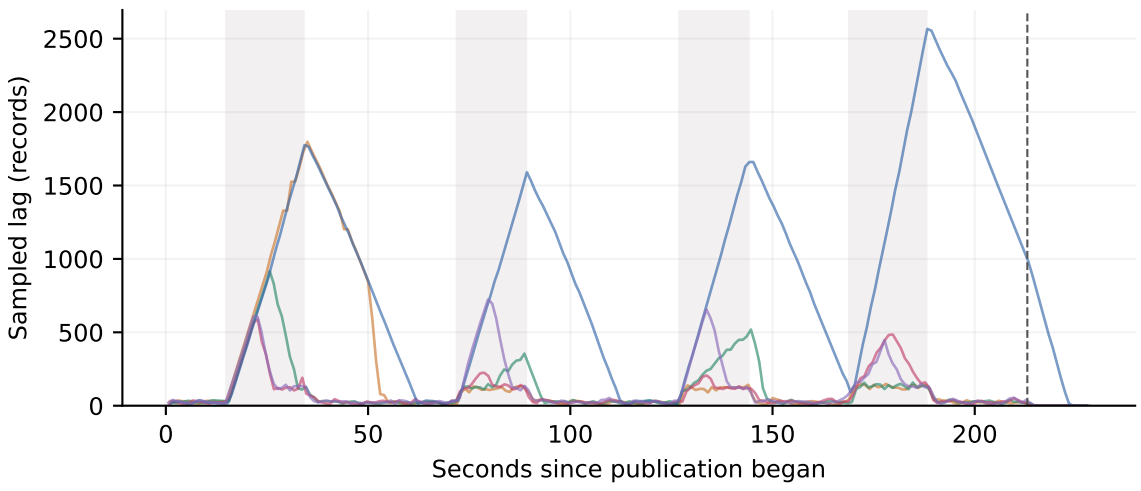


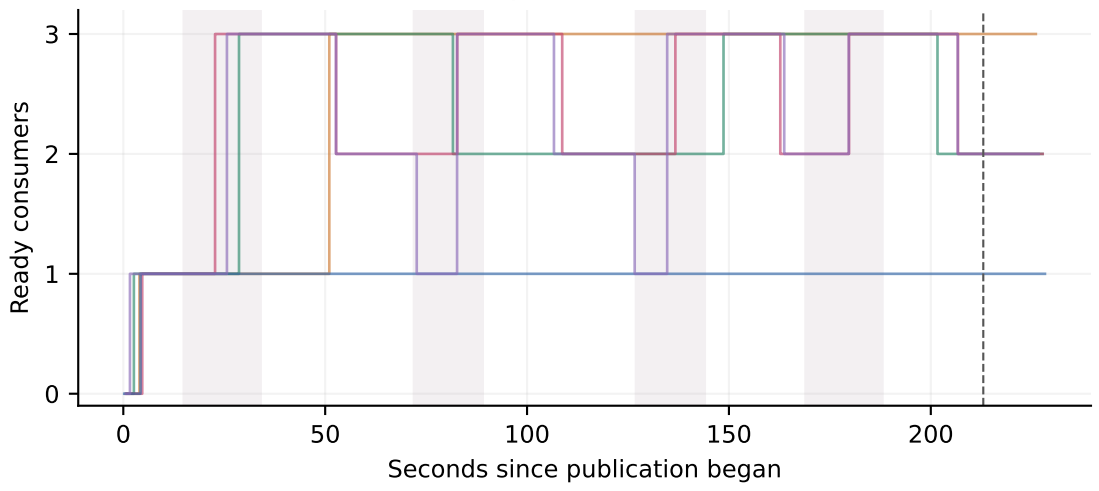
Ramp





Repeated Sales





Steady

